

M38 — Anthropic key live-refresh on station turn + friendlier 401

Phase: Phase 7 follow-on (post-M37, operator-reported bug fix) **Status:** **DONE** **Blocked by:** M35 (Engage flow exposed the bug) **Blocks:** none **Estimated effort:** 0.5 day

1. Purpose

Operator-reported after M37:

"Respond by saying one sec, then this error message — System: AuthenticationError: Error code: 401 — {type: 'error', 'error': {type: 'authentication_error', 'message': 'invalid x-api-key'}, 'request_id': '...'}"

Three problems wrapped in one report:

1. **Stale** `sess.api_key`. The encounter's Anthropic key is snapshotted at `/api/room/start` time from `vault.get("ANTHROPIC_API_KEY")`. If the operator later rotates the key (or the vault key was wrong all along), the snapshot is invalid and every Claude call from the station-turn route 401s. The operator's only previous fix was to end the room and start a new one — disruptive mid-session.
2. **No validation at room start**. When the vault has no Anthropic key, `/api/room/start` still succeeds (creating an encounter with `api_key=""`). The bug surfaces only on the first turn — far from the cause.
3. **Hostile error message**. The raw Python repr of the Anthropic 401 exception (`"AuthenticationError: Error code: 401 - {...}"`) bubbles up into the chat verbatim. Doesn't tell the operator where to fix it.

The fix mirrors the ElevenLabs `_runtime_key` pattern already in `portal/voices.py`: a process-wide cache that any operator-auth route stamps when it reads the key from vault. The station-turn route prefers the cache over `sess.api_key`, so updating `/portal/credentials` propagates to live encounters without a restart.

2. Structure

Files touched:

- `portal/server.py`:
- New module-level `_anthropic_runtime_key`: `str = ""`.
- New helper `_capture_anthropic_key(key)` — updates the cache.

Empty strings never overwrite a previously-cached non-empty value (protects against accidental clears). - New helper `_resolve_anthropic_key(sess)` — returns the cache if set, otherwise `sess.api_key`. This is what the station-turn route uses to pick the best-known key. - Hooks: `_capture_anthropic_key(...)` is called from `/portal/credentials` (POST), `/portal/control/start` (single- patient room start), and `/api/room/start` (multi-patient room start). Each of those reads the vault key; the cache reflects that read. - `/api/room/start` now raises **400** with a clear message when the vault has no `ANTHROPIC_API_KEY`. Prevents the "filler plays, then 401 in chat" experience that triggered M38. - `/api/station/{join_code}/{station_id}/turn` now uses `_resolve_anthropic_key(sess)` instead of `sess.api_key` directly. It also keeps `sess.api_key` in sync with the cache (so other paths benefit). When the key is empty, returns a structured `{"ok": False, "error": "No Anthropic API key configured. Open /portal/credentials ..."}.` - The Claude call inside the same route is wrapped in `try / except Exception` — a thrown `AuthenticationError` (401) or runtime error is caught and translated into a chat-friendly string that points at `/portal/credentials`.

`runtime.take_turn`'s own `{"ok": False, "error": ...}` return shape is also inspected for 401-shaped substrings and rewritten.

No schema migration. No dataclass change. The encounter still carries `api_key` (the snapshot); we just don't blindly trust it.

3. Uses

3.1 Operator flow (the bug path that prompted the fix)

1. Operator stored an Anthropic key in `/portal/credentials` weeks ago. That key was later rotated (Anthropic admin console). The operator forgot to update the vault. 2. Operator starts a multi-patient room → encounters now carry `sess.api_key = "<stale key>"`. 3. Operator clicks Engage on the encounter console → lands on `/station/{join}/INST-{persona}`. 4. Operator presses PTT, speaks → STT result → POST `/turn`. 5. Filler audio plays through ElevenLabs (different key path — that one works). 6. Server calls `runtime.take_turn(...)` with the stale key → Anthropic returns 401. 7. **Before M38:** chat shows `"System: AuthenticationError: Error code: 401 - {...raw repr...}"` and the operator has no idea where to fix it. 8. **After M38:** chat shows `"Anthropic rejected the API key (401). Update ANTHROPIC_API_KEY at /portal/credentials and try again — the new key applies immediately."`

3.2 Operator flow (the recovery path)

1. Operator sees the friendly 401 chat error. 2. Operator opens `/portal/credentials` in a new tab, pastes the fresh Anthropic key, clicks Save → POST `/portal/credentials`. 3. The route's `_capture_anthropic_key(value.strip())` updates the process-wide cache. 4. Operator returns to the engage chat tab, presses PTT again. 5. `/api/station/.../turn` reads the cache via `_resolve_anthropic_key(sess)` → gets the new key. 6. Claude call succeeds. Operator is back in business — no room restart.

3.3 The cache-precedence rules

The cache is preferred for one reason: it reflects the latest operator-confirmed key value. The encounter's snapshot is the fallback for two cases:

1. **Fresh process, no operator route has read vault yet.** Cache is empty, snapshot is whatever room-start stamped. Use snapshot. 2. **Cache somehow got cleared** (defensive). We protect against `_capture_anthropic_key("")` overwriting the cache, but in case of future code paths that clear it, the snapshot is the lifeboat.

`_resolve_anthropic_key(sess)` returns whichever exists; only when BOTH are empty does the route return the friendly 503-style fallback message.

4. Functions (exported API surface)

Symbol	Where	Purpose
<code>_anthropic_runtime_key</code> (module var)	<code>portal/server.py</code>	Process-wide cache of the latest Anthropic key seen by an operator-auth route.
<code>_capture_anthropic_key(key)</code>	<code>portal/server.py</code>	Update the cache. Empty strings are no-ops.
<code>_resolve_anthropic_key(sess)</code>	<code>portal/server.py</code>	Best-known key for a station-turn callsite: cache → session snapshot → "".

Symbol	Where	Purpose
(refresh hooks)	/portal/credentials POST, /portal/control/start, /api/room/start	Each calls <code>_capture_anthropic_key</code> after resolving the vault key.
(translation)	/api/station/{join}/{station}/turn	Catches 401 / authentication errors and returns a friendly chat message.

5. Limitations

- **The cache is per-process and lost on restart.** A server restart

must read the key from vault to repopulate. The first operator route after restart does that.

- `_capture_anthropic_key("")` **is a no-op.** If the operator

deletes the key from `/portal/credentials`, the cache retains the prior value until the process restarts. This is intentional — better to keep working with the prior key than to immediately break every live encounter. The OPERATOR can sidestep this by setting the key to an obviously-invalid value (e.g. "sk-ant-x"), but the normal flow is to update to a valid key.

- **No test for the catch-401-during-turn path.** That would need

Anthropic credentials and a real 401 response. Verified by inspection. The friendly "no key configured" path IS tested.

- ****The first turn after room start may still use the stale**

snapshot** if the operator has NOT visited any cache-priming route in the same process lifetime.

`/api/room/start` itself is a priming route, so this is essentially never an issue: the room start call seeds the cache.

- **ElevenLabs key path was already correct** (it has its own

`_runtime_key` in `voices.py`). M38 only touches the Anthropic side. The two caches are independent — symmetric design.

- **The friendly 401 message hard-codes "/portal/credentials".** If

the route ever moves, the message goes stale. Acceptable risk.

6. Test status

Test file	Asserts	Status	Last run
tests/v7/test_anthropic_key_live_refresh.py::test_room_start_400s_when_no_anthropic_key_in_vault	Missing vault key → 400 with <code>/portal/credentials</code> hint	PASS	2026-05-27
...::test_room_start_seeds_anthropic_runtime_cache	Successful room-start seeds the cache	PASS	2026-05-27
...::test_credentials_post_refreshes_anthropic_runtime_cache	<code>/portal/credentials</code> update refreshes the cache live	PASS	2026-05-27
...::test_resolve_anthropic_key_prefers_runtime_cache_over_snapshot	Cache wins when both are set	PASS	2026-05-27

Test file	Asserts	Status	Last run
...::test_resolve_anthropic_key_falls_back_to_session_snapshot	Empty cache falls back to snapshot	PASS	2026-05-27
...::test_capture_does_not_overwrite_cache_with_empty	Empty_capture_anthropic_key("") is a no-op	PASS	2026-05-27
...::test_station_turn_returns_friendly_message_when_no_key	Empty cache + empty snapshot → friendly 200 JSON, not a raise	PASS	2026-05-27
Full v7 suite	250 passed, 1 skipped (M20 Playwright skip — unchanged)	PASS	2026-05-27

7. Change list

Date	Author	Change	Files
2026-05-27	claude-code	Initial M38 implementation: process-wide Anthropic key cache + 3 refresh hooks + fail-fast room start + friendly 401 in station turn; 7 new tests	portal/server.py, tests/v7/test_anthropic_key_live_refresh.py (new)

8. Open questions / known issues

- **Should the operator turn route also use `_resolve_anthropic_key`?**

It currently does the right thing in a slightly different way: `live_key = vault.get("ANTHROPIC_API_KEY")` or `sess.api_key`. Since the operator route has vault access, it can read fresh on every call — no need for the cache there. Acceptable redundancy.

- **Cache invalidation when the key is deleted.** Currently we

preserve the old cached value. If the operator wants to FORCE all live encounters to fail (e.g. wind down a classroom), they need to restart the process. A future M39 could add an explicit "purge cache" route gated behind `require_instructor`.

- ****The friendly message in `/api/station/.../turn` checks**

substrings like "401", "invalid x-api-key", "authentication".** If Anthropic changes the wording, the friendly translation silently degrades to the generic "Turn failed: ...". Tracked.

- ****Validation at room start could also test the key with a**

cheap probe call** (the existing `/portal/credentials/test` endpoint does this for the same key). Skipped here to keep `/api/room/start` latency low — the friendly station-turn 401 is the safety net.

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